Mohaddeseh (Sahar) Hosseinzadeh

Computational Neuroscience researcher | Machine learning Engineer | Software developer

Check out my website: https://synapticsahar.github.io   SynapticSahar   LinkedIn

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Interests: Computational Neuroscience | Machine Learning | Biomedical Signal Processing | Medical AI | Product Development

EXPERIENCE

ML Engineer — Incoming Intern
Barcelona Supercomputing Center — *Jul 16 – Sep 30, 2026*
- Joining the Digital Health Unit for a summer internship under the supervision of Paula Petrone, focusing on developing AI for healthcare.

Researcher — Brain Metastability and MOP
CNS Lab, TCN Lab, UPF — *Sep 2025 – Present*
- ***Core question:*** Does the dynamic structure of brain state transitions shape individual differences in cognition and decision-making?
- Computing whole-brain metastability via vortex-space and clustering, measuring Shannon and path entropy per subject, and correlating with cognitive flexibility and behavioral performance.
- Modeling maximum ocupancy principle and measuring its performance by comparing predicted transition probabilities to empirical ones
- *Stack:* fMRI, HCP, Vortex Space, Kuramoto, Entropy, k-means, Ridge Regression, MATLAB

Researcher — Sharp Wave Ripple Detection from LFP
CMC Lab — *Apr 2025 – Feb 2026*
- Built full SWR detection pipeline without manual threshold tuning :NMF-based spectral feature learning, Transformer-based modeling, and session-level evaluation
- *Stack:* PyTorch, Transformers, NMF, scikit-learn, NumPy

ML Engineer — EEG Emotion Recognition
Imam Khomeini Hospital · Clinical Lab — *Feb – Aug 2025*
- Built full EEG emotion recognition pipeline: bandpass filtering, ICA artifact removal, epoching on clinical EEG using MNE-Python. Trained EEGNet for 4-class emotion detection (valence/arousal quadrants) from frequency band power features (theta, alpha, beta, gamma). Deployed real-time inference: FastAPI backend + Streamlit frontend. Dockerized on AWS EC2.
- *Stack:* MNE-Python, EEGNet, PyTorch, FastAPI, Streamlit, Docker, AWS EC2

ML Engineer — Deep Learning on EEG/EMG — Wearable BCI
Myant Inc. ,Canada — *Jan 2024 – Jan 2025*
- Preprocessed EEG/EMG from textile wearables; built eye tracking and sleep staging models from a headband; developed RNN for EMG-based AR/VR movement steering using a glove dataset.

Full-Stack Software Development
Rahnema College — *Mar 2021 – Mar 2022*
- Built and deployed a full-stack web app end-to-end.

PROJECTS

Deep Q-Network for the Beer Game
Bachelor's Thesis, 2023
- PyTorch · DQN · Multi-agent RL — extended DQN for multi-agent inventory control; tested under varying demand conditions.
  Code: Github

EEG → Image Decoding via CLIP (THINGS-EEG2)
2025
- Transformer encoder aligns EEG to CLIP embeddings via contrastive loss (NT-Xent); retrieves top-3 image categories the brain saw. GPT-4o narrates the decoded perception.
- *Stack:* CLIP, Contrastive Learning, Transformers, MNE-Python, HuggingFace, OpenAI API

Seizure Detection — Fine-Tuned Foundation Model
2025
- Fine-tuned PatchTST on CHB-MIT Scalp EEG (PhysioNet); benchmarked against clinical baselines. Deployed via FastAPI + Streamlit on AWS SageMaker.
- *Stack:* PatchTST, HuggingFace, Clinical EEG, FastAPI, AWS SageMaker

Synaptic Plasticity Modelling in Mouse V1
Sep 2024 – Apr 2025
- Brian2 · STDP · Allen Institute Ophys Dataset — simulated reward-modulated STDP in SNNs; studied VIP/SST/excitatory interactions under rewarded vs. passive conditions.
  Code: Github

Shannon Entropy for fMRI Task Prediction (GRU)
July 2024
- PyTorch · HCP · Transfer Learning — GRU with cross-subject transfer; entropy features outperformed raw BOLD across all runs.
  Code: Github   *Cert:* Here   *Talk:* Here

3D U-Net for Brain Tumor Segmentation
Feb 2024
- PyTorch · BraTS2020 · Dice loss — 3D U-Net for volumetric MRI segmentation with augmentation and LR scheduling.
  Code: Github

EDUCATION

M.Sc., Computational Neuroscience
Pompeu Fabra University — Sep 2025 – Sep 2026

B.Sc., Industrial Engineering
Sharif University of Technology — 2017 – 2023
- Teaching Assistant: Linear Algebra (40+ undergrads)

SKILLS

- **Languages & Scripting**
  Python, MATLAB, C/C++, R, Java, JavaScript, Shell
- **ML / Deep Learning**
  PyTorch, TensorFlow, Keras, scikit-learn, SciPy
  CNNs, RNNs/GRU, Transformers, RL, SNNs, DQN
- **Signal Processing**
  LFP, EEG, EMG, fMRI
- **LLMs & GenAI**
  HuggingFace, LangChain, RAG, OpenAI API, PEFT/LoRA, Prompt Engineering
- **Neuro Tools**
  Allen SDK, NEST, Brian2, PyNN, NEURON
- **Cloud / HPC**
  AWS (EC2, SageMaker, Lambda, Lightsail), Linux
- **Software Practices**
  Clean Code, DRY, KISS, YAGNI, SOLID, OOP, Design Patterns
  Git, REST APIs, CI/CD, Docker, Agile/Scrum

INTERNSHIPS

Computational Neuroscience
Neuromatch Academy — *July 2024*
- *Certificate:* Here

REFERENCES

Paula Petrone	Barcelona supercomputing center
Dr. Ruben Moreno-Bote	TCN Lab, UPF
Prof. Gustavo Deco	CNS Lab, UPF
Dr. Shervin Safavi	CMC Lab, TU Dresden
Dr. Milad Alizadeh-Meghrazi	Myant Inc.
Alexandre Hyafil	CRM, Barcelona
Klaus Wimmer	Wimmer Lab, CRM